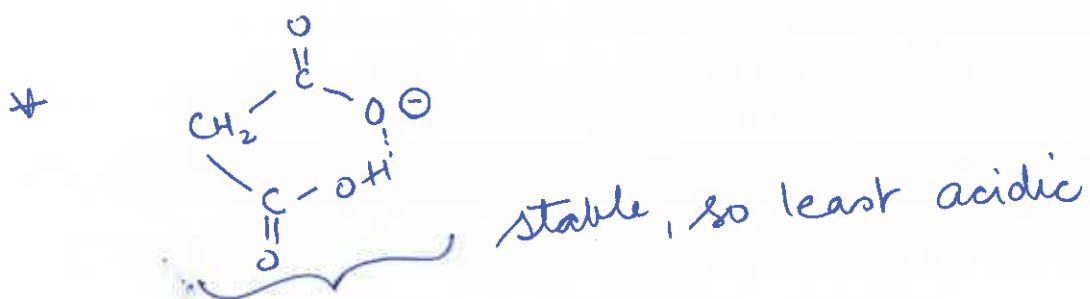
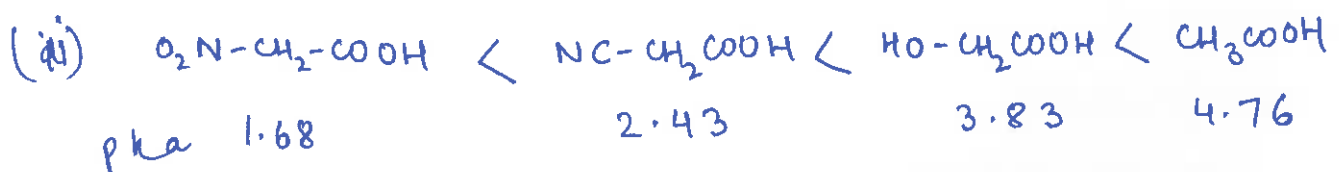
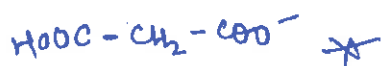
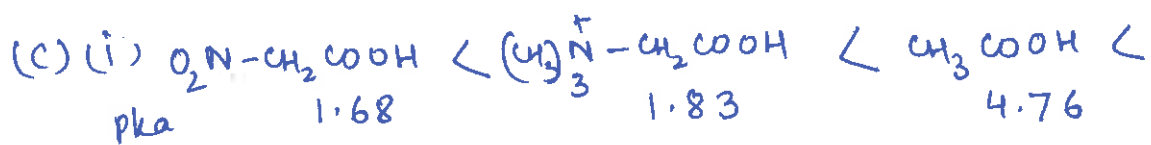
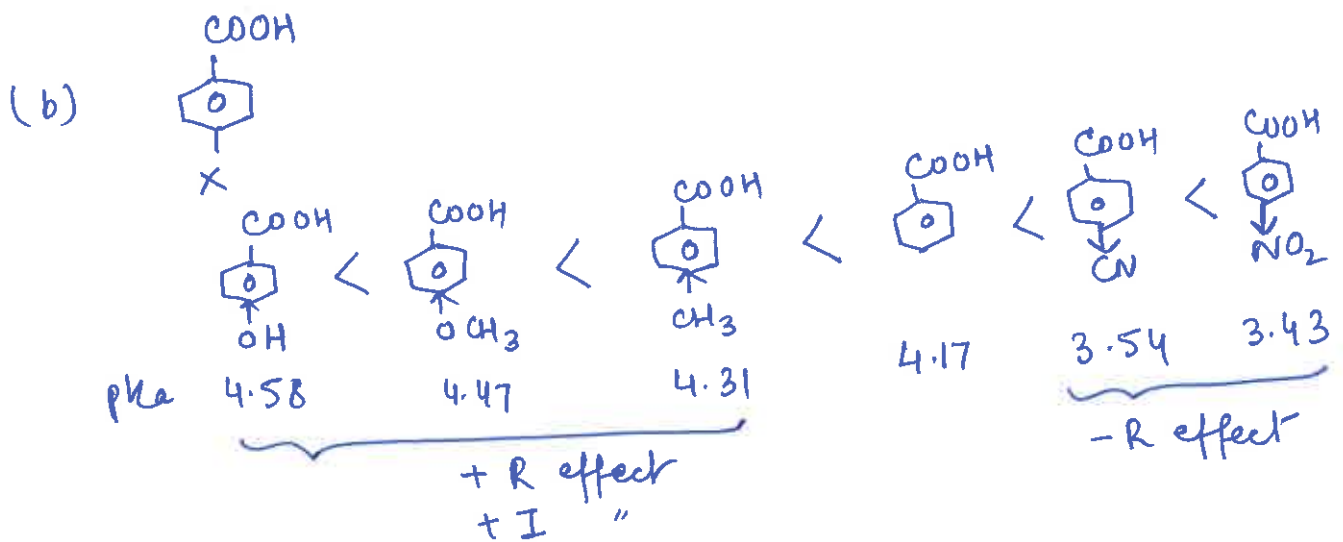
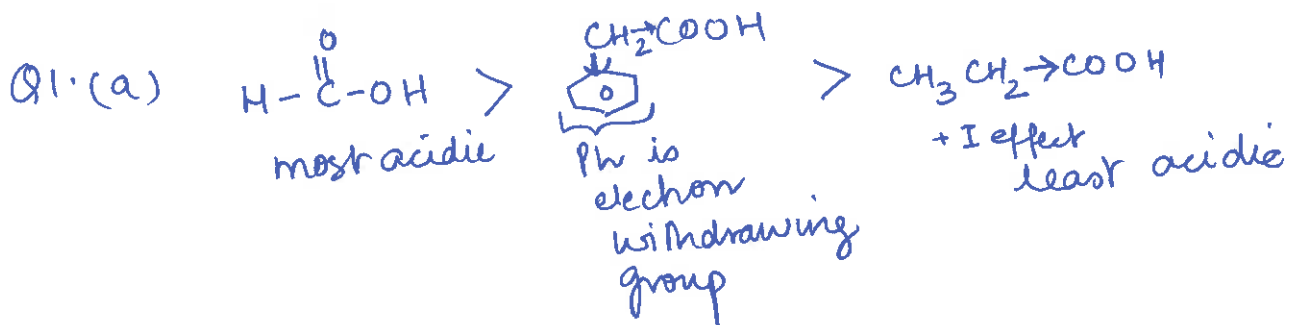
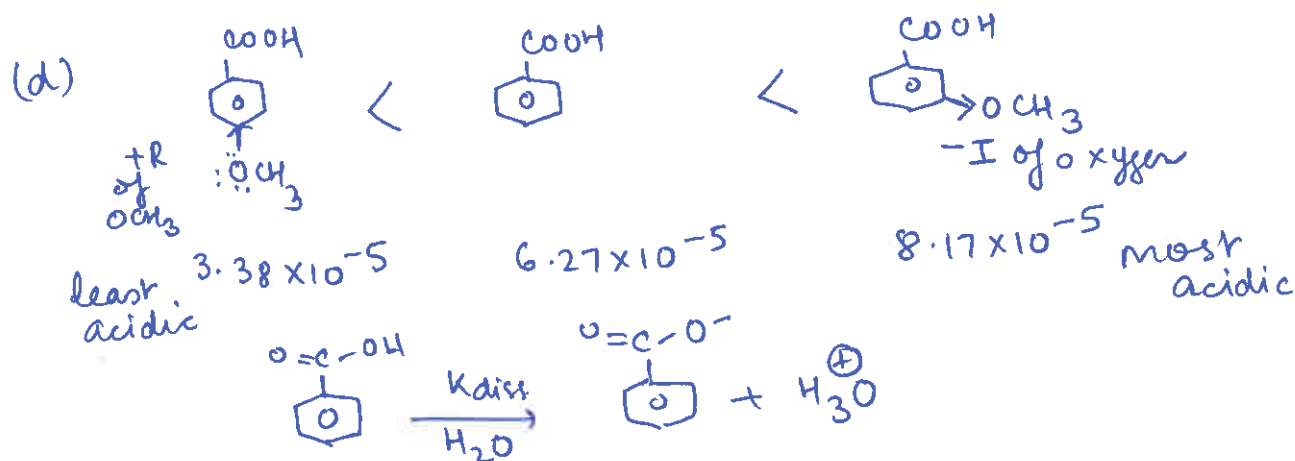


ACML 100 Tutorial 3 Answers

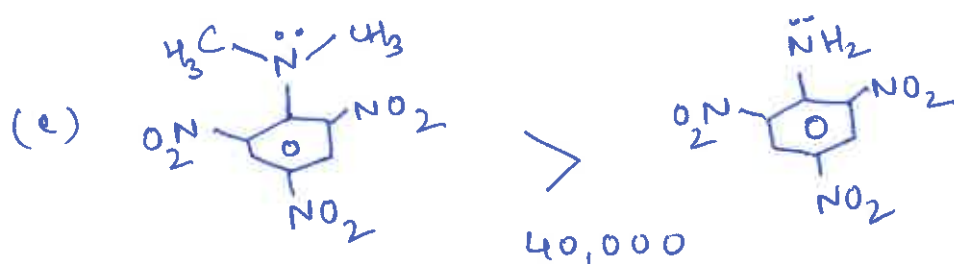
①



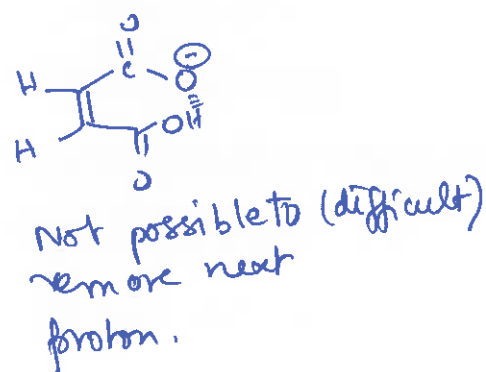
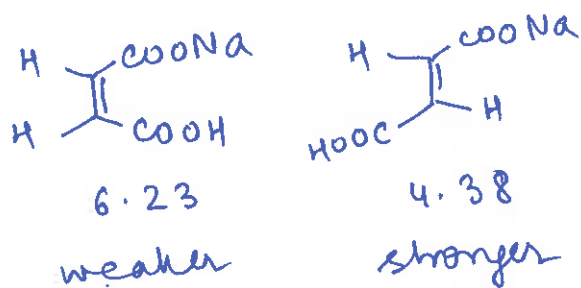
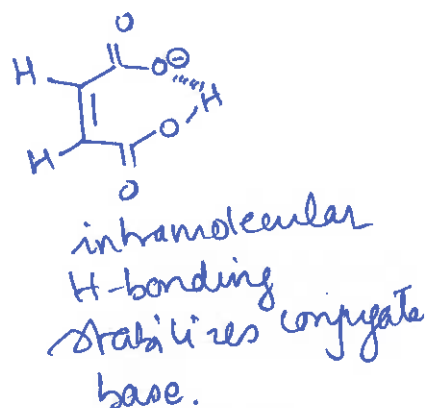
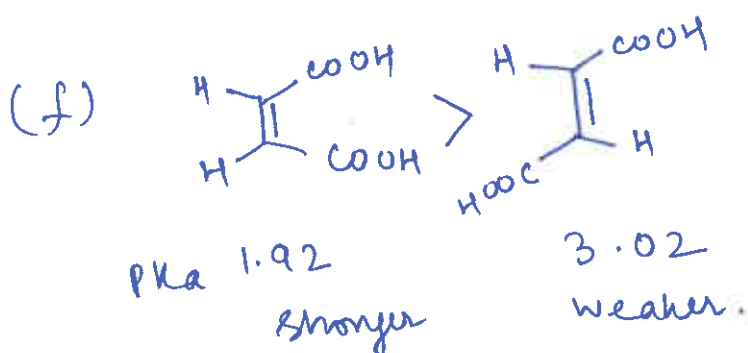


Higher K_{diss} ; more acidic

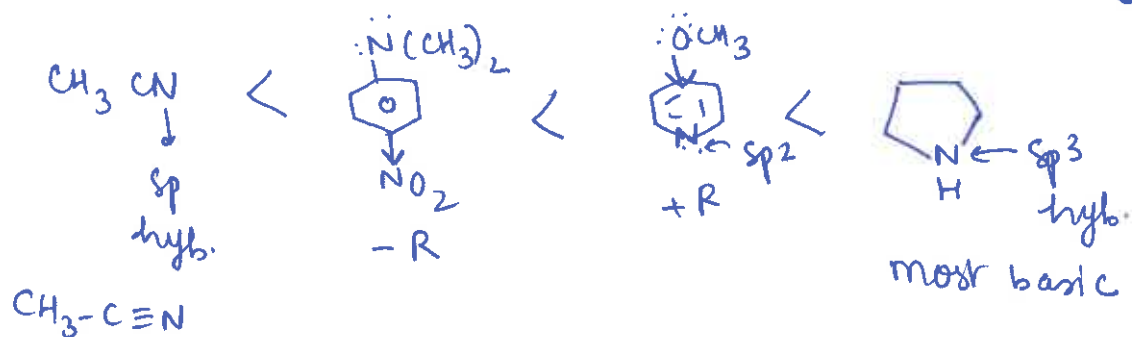
Reason: above



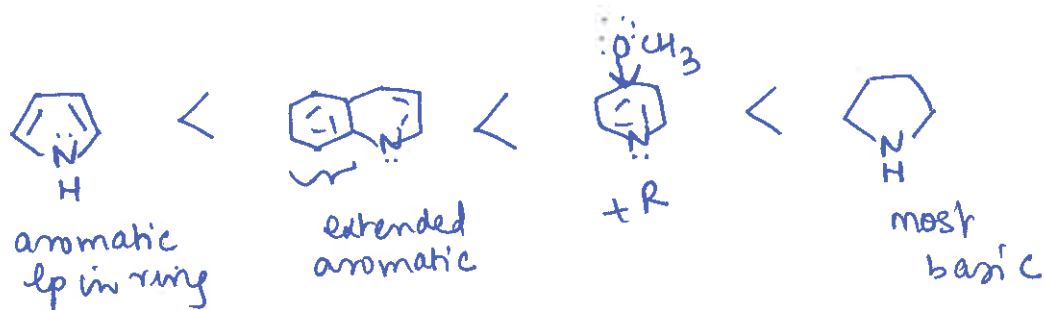
Reason: lone pair of electrons are not in plane of ring. $\therefore$ cannot get delocalized over ring because of steric bulk of $N(CH_3)_2$ gp.



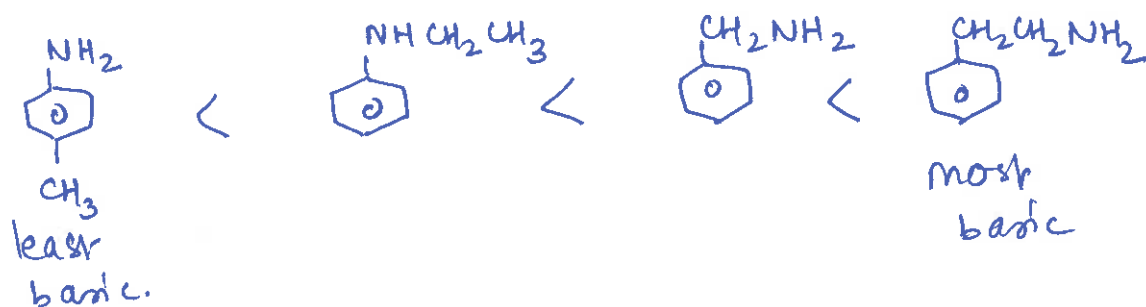
Q2. a)



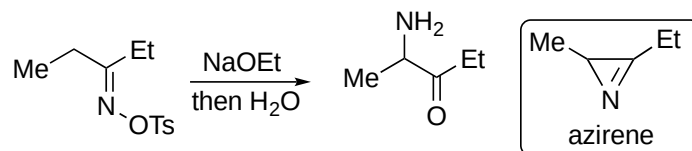
b)



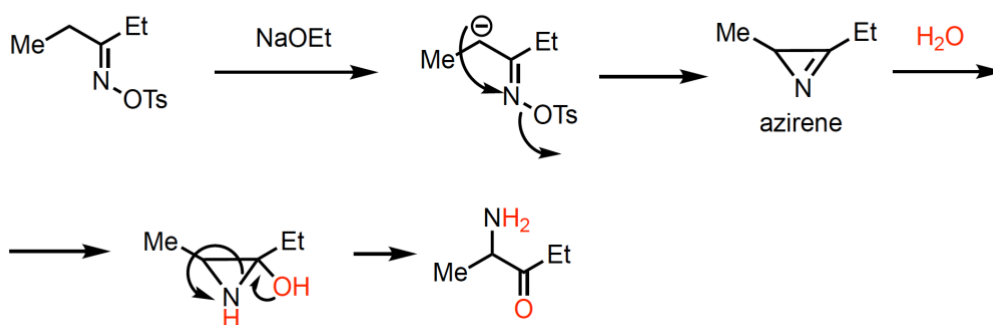
c)



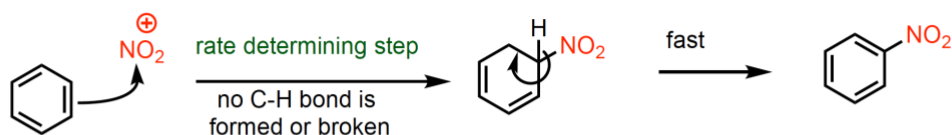
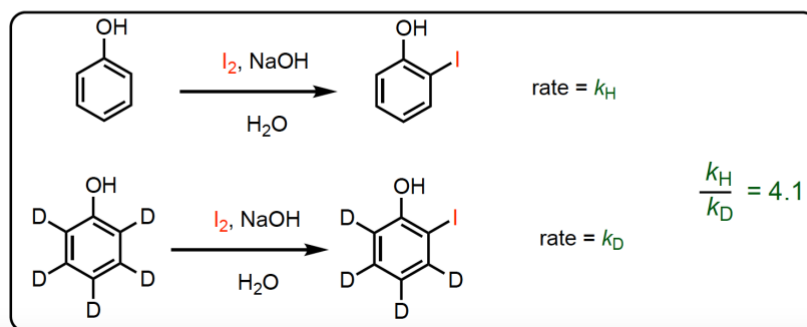
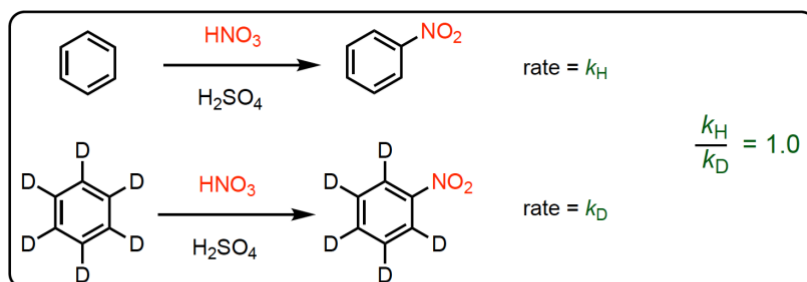
3. For the following transformation, intermediate azirene was isolated. With the help of intermediate structure, propose a suitable mechanism for this reaction.



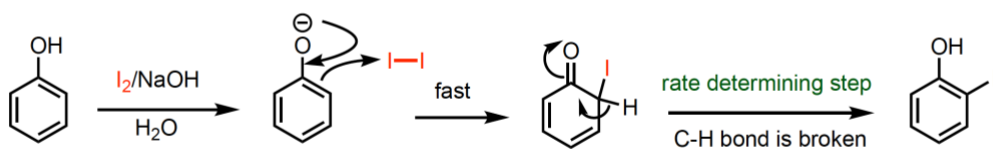
Neber rearrangement:



4. Explain the mechanism of the following transformation and identify the rate determining step.



replacing the hydrogen atoms round the benzene ring with deuteriums, The rate of the reaction stays the same.



the reaction is the iodination of phenol in basic solution, there is a deuterium isotope effect of $k_{\text{H}}/k_{\text{D}} = 4.1$. Clearly, the second step must be the rate-determining step—the phenolate ion reacts so rapidly that the first step is faster than the second